Deep Learning and Deterministic Algorithms for Document Verification

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1. Abstract

We present a robust verification engine pairing DP token alignment with visual differencing.
Across benchmark tests on multi-source documents, our model yields an empirical accuracy of 72.1%.

2. Introduction

Document tampering represents a substantial threat across academic admissions and audit workflows.
Traditional text-only extraction frequently ignores micro-typographic shifts, font changes, and layout variance.

3. Methodology

Our architecture combines the open-source pdf-diff bounding box tracker with PyMuPDF layout analysis.
Computer vision differencing with OpenCV captures altered diagram contours and embedded imagery.
An additional experimental paragraph was appended to Methodology in the revision.

4. Results

Rigorous empirical evaluation demonstrates measured precision of 72.1% with sub-second latency.

5. Conclusion

The integrated pipeline effectively pinpoints both typographic variance and visual document modifications.